

# Local AI Model Hosting Guide

No-Docker Implementation for Secondary Drive Optimization

This document provides a comprehensive setup for hosting local Large Language Models (LLMs) such as **Qwen2.5-Coder** using **Ollama** and **Open-WebUI**. It is designed for environments where Docker is restricted and optimized for storage on a secondary drive (A:).



## Prerequisites

- **Ollama:** Native Windows installation.
- **Python:** Version 3.11 or higher (with PATH configured).
- **Storage:** Dedicated secondary partition (A:) for model weights and environment files.



## Installation & Setup

### 1. Engine Installation (Ollama)

1. Run the `ollamaSetup.exe` installer.
2. **Important:** Right-click the Ollama icon in the system tray and select **Quit** before proceeding to configuration.

### 2. Drive & Environment Configuration

Redirecting data to the A: drive prevents C: drive congestion.

1. Open "Edit the system environment variables" in Windows.
2. Create the following User Variables:

| Variable Name              | Variable Value                 | Purpose                               |
|----------------------------|--------------------------------|---------------------------------------|
| <code>OLLAMA_MODELS</code> | <code>A:\ollama_storage</code> | Redirects LLM weights to A: drive     |
| <code>DATA_DIR</code>      | <code>A:\webui_data</code>     | Redirects chat history/DB to A: drive |

Create the folders on your A: drive:

```
mkdir A:\ollama_storage  
mkdir A:\webui_data
```

### 3. Native Web-UI Setup

Install the interface into a Python virtual environment on the secondary drive.

#### 1. Navigate and Initialize:

```
cd /d A:  
mkdir local_ai_project  
cd local_ai_project
```

#### 2. Create/Activate Virtual Environment:

```
python -m venv ai_env  
ai_env\Scripts activate
```

#### 3. Install Package:

```
pip install open-webui
```

### 4. Verification & Launch

1. Restart Ollama from the Start menu.
2. Pull the optimized coding model:

```
ollama pull qwen2.5-coder
```

3. Launch the server with network visibility:

```
open-webui serve --host 0.0.0.0 --port 8080
```

## Issue Summary & Solutions

| Issue                  | Solution                                                                                 |
|------------------------|------------------------------------------------------------------------------------------|
| No Docker Support      | Native Python installation using <code>pip</code> and <code>venv</code> .                |
| C: Drive Storage Bloat | Environment variable redirection ( <code>OLLAMA_MODELS</code> & <code>DATA_DIR</code> ). |
| Mobile Access Blocked  | Hosting on <code>0.0.0.0</code> and adjusting Windows Firewall for port 8080.            |

### Pro-Tip: Automation

Create a `launch.bat` file for one-click startup:

```
@echo off
SET OLLAMA_MODELS=A:\ollama_storage
SET DATA_DIR=A:\webui_data
start /b ollama serve
call A:\local_ai_project i_env\Scripts ctivate
open-webui serve --host 0.0.0.0 --port 8080
```